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TS31

# Title Options for Consultation

Five Abstracts From Our Earlier Proposals

These are five titles from our earlier proposals that the group would like to discuss. All five use public data, so under the rule on open-source data each one needs a private client or a different data source. We would like to ask which of them can still be pursued and what we would need to change.

### Title 1: Forecasting Luzon Grid Electricity Demand From Heat Index Using LightGBM

Electricity demand in Luzon rises on hot days, and the grid is dispatched in five-minute intervals. This study forecasts regional power demand from heat index readings and calendar features using LightGBM, and compares it with a simple seasonal baseline. The demand records come from the 5-minute regional files that IEMOP publishes. The expected output is a day-ahead demand forecast with an error report for hot days and normal days.

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| **Data** | IEMOP 5-minute regional demand files (public). Heat index source still to be confirmed. |
| **Size** | About 60,000 Luzon intervals from March 2026 onward, and growing. Files before February 2026 returned nothing. |
| **Method** | LightGBM against a seasonal baseline. |
| **SDG** | SDG 7, proposed, to confirm with the adviser. |
| **Client** | No client yet. A private heavy electricity user, such as a factory or a mall, is the kind to look for. |

### Title 2: Detecting Red Flags in Philippine Government Procurement Awards Using Benford's Law and Network Analysis

Government procurement involves very large sums, and checking millions of award records by hand is not realistic. This study screens PhilGEPS contract awards for red flags using Benford's Law on contract amounts and network analysis of how agencies and suppliers are linked. A first look across the full 2013 to 2025 data shows the digit 9 leading in 7.8 percent of awards where Benford's Law expects about 4.6 percent, and 5.5 percent of awards dated before they were published. A red flag is a reason to look closer and not proof of wrongdoing. The expected output is a ranked list of agencies and suppliers worth a closer look, with the checks that flagged them.

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| **Data** | PhilGEPS contract awards, compiled in an open dataset on GitHub (MIT license). |
| **Size** | 2,769,747 awards from 2021 to 2025 by award date. 5,481,161 awards in the full 2013 to 2025 set. |

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| **Method** | Benford's Law first-digit test and network analysis of agency and supplier links. |
| **SDG** | SDG 16, targets 16.5 and 16.6. |
| **Client** | No client yet. A private watchdog, audit or compliance group, or a civic tech group, is the kind to look for. |

### Title 3: Profiling Philippine Air Cargo Routes Using K-Means Clustering

The Philippine Statistics Authority records air cargo value by origin station, destination station and commodity. This study groups routes by what they carry and how much value they move, using K-Means clustering, so that similar routes can be compared. While checking the table we found that the 1997 values are an exact copy of the 1995 values, so those years will be treated as unreliable. The expected output is a set of route profiles with a short description of each cluster.

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| **Data** | PSA OpenSTAT air cargo table by commodity, origin station and destination station (public). |
| **Size** | 10,099 route and commodity flows with a recorded value from 2021 to 2025, 7,767 of them non-food. The full 1995 to 2025 table has 147,932. |
| **Method** | K-Means clustering of route profiles. |
| **SDG** | SDG 9, proposed, to confirm with the adviser. |
| **Client** | No client yet. A freight forwarder or cargo operator that shares shipment records is the kind to look for, with the PSA table as the national benchmark. |

### Title 4: Assessing Cause of Death Data Quality Across Philippine Regions Using Hierarchical Clustering

Cause of death statistics are only useful when the cause is clearly recorded. In the 2024 PSA table, ill-defined causes make up 4.7 percent of deaths nationally, 12.5 percent in BARMM and 1.9 percent in NCR. This study uses hierarchical clustering to group regions by how their causes of death are recorded and to show where the recording looks weak. It is the smallest of the five titles and the data is aggregated by region and cause, so we would like to ask whether it can count as record-level data.

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| **Data** | PSA OpenSTAT deaths by cause and region, 2024 (public). |
| **Size** | 31,491 detailed records for 2024. Aggregated, with only two years. |
| **Method** | Hierarchical clustering (Ward). |
| **SDG** | SDG 3, proposed, to confirm with the adviser. |
| **Client** | No client yet, and none is easy to find. A hospital or an insurer would be a stretch. |

**Title 5: Forecasting Benguet Vegetable Prices Using XGBoost**

Vegetable prices in Benguet change quickly, and farmers and traders plan around them. This study forecasts short-term prices of selected highland vegetables using XGBoost and compares it with a simple baseline. BAPTC declined to share its raw data in September 2026, so the data would have to come from DA Bantay Presyo bulletins or from a client that keeps its own sales records. The expected output is a weekly price forecast with an error report.

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| **Data** | DA Bantay Presyo bulletins (public), or a client's own sales records. |
| **Size** | Not counted yet. No raw dataset is secured. |
| **Method** | XGBoost against a simple baseline. |
| **SDG** | SDG 2, proposed, to confirm with the adviser. |
| **Client** | No client yet. A vegetable trader or cooperative that keeps its own sales records is the kind to look for. |

*Prepared with AI assistance (Claude) and to be reworded by the group. The PhilGEPS and air cargo counts were checked on 2026-10-16 and the other figures on 2026-10-04.*